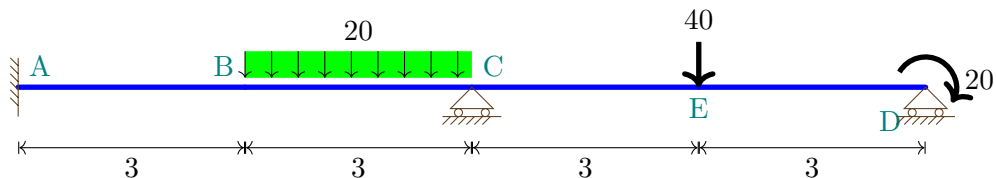


ENCI335
Structural Analysis and Systems I
Assignment 2 Solution

Chin-Long Lee

Question 1 Integration Method for Beam Analysis

Consider the following beam with a uniform flexural rigidity EI , use **the integration method** to solve for its reactions and draw the BMD, SFD, and deflected shape.



Analysis Guide:

- Start with the equation $\frac{dV}{dx} = w(x)$, and use Macaulay brackets.
- The reactions at the end supports (points A and D) and the external point moment at point D should not be included in the load intensity function $w(x)$.
- The point moment at point D is part of the boundary conditions and results in a hogging moment.
- Three simultaneous equations may be needed to solve for three unknowns. Feel free to use Python to find the solution.
- Answers for checking: $R_A = 12.14$, $M_A = 20.54$, $R_C = 70.8$, the bending moment at point C is -37.7 (hogging), the shear force at point B is 12.14 , and the rotation at point C is $5.36/EI$.

Solution:

We can start with the following equation:

$$\frac{dV}{dx} = -20\langle x-3 \rangle^0 + 20\langle x-6 \rangle^0 + R_C\langle x-6 \rangle^{-1} - 40\langle x-9 \rangle^{-1} \quad (1)$$

with x starting from the left end and R_C being the upward reaction force at C. Note that we don't need to include the forces and moments on the boundaries in the load function. Integrating the equation once and twice gives the shear force $V(x)$ and bending moment $M(x)$, respectively:

$$V(x) = -20\langle x-3 \rangle^1 + 20\langle x-6 \rangle^1 + R_C\langle x-6 \rangle^0 - 40\langle x-9 \rangle^0 + C_1 \quad (2)$$

$$M(x) = -10\langle x-3 \rangle^2 + 10\langle x-6 \rangle^2 + R_C\langle x-6 \rangle^1 - 40\langle x-9 \rangle^1 + C_1x + C_2 \quad (3)$$

Knowing that the moment at the right end where $x = 12$ is 20 (hogging), we can write:

$$M(12) = -20 = -10(9)^2 + 10(6)^2 + R_C(6) - 40(3) + 12C_1 + C_2 \quad (4)$$

or

$$6R_C + 12C_1 + C_2 = 550 \quad (5)$$

This equation gives a relationship among R_C , C_1 , and C_2 that will be used later.

Given a constant EI , integrating the equation for $M(x)$ once and twice gives the rotation and deflection, respectively:

$$EI\theta(x) = -\frac{10}{3}\langle x-3 \rangle^3 + \frac{10}{3}\langle x-6 \rangle^3 + \frac{R_C}{2}\langle x-6 \rangle^2 - 20\langle x-9 \rangle^2 + \frac{C_1}{2}x^2 + C_2x + C_3 \quad (6)$$

$$EIv(x) = -\frac{5}{6}\langle x-3 \rangle^4 + \frac{5}{6}\langle x-6 \rangle^4 + \frac{R_C}{6}\langle x-6 \rangle^3 - \frac{20}{3}\langle x-9 \rangle^3 + \frac{C_1}{6}x^3 + \frac{C_2}{2}x^2 + C_3x + C_4 \quad (7)$$

Since the deflection and rotation at A ($x = 0$) are zero, we can conclude that $C_3 = C_4 = 0$. Because the deflection is also zero at C ($x = 6$) and D ($x = 12$), we have:

$$v_C = 0 = -\frac{5}{6}(3)^4 + \frac{C_1}{6}(6)^3 + \frac{C_2}{2}(6)^2 \quad (8)$$

$$v_D = 0 = -\frac{5}{6}(9)^4 + \frac{5}{6}(6)^4 + \frac{R_C}{6}(6)^3 - \frac{20}{3}(3)^3 + \frac{C_1}{6}(12)^3 + \frac{C_2}{2}(12)^2 \quad (9)$$

Therefore, together with the earlier equation, we have the system:

$$6R_C + 12C_1 + C_2 = 550 \quad (10)$$

$$36C_1 + 18C_2 = 67.5 \quad (11)$$

$$36R_C + 288C_1 + 72C_2 = 4567.5 \quad (12)$$

To solve this system of simultaneous equations, we can rewrite it as the following matrix equation:

$$\begin{bmatrix} 6 & 12 & 1 \\ 0 & 36 & 18 \\ 36 & 288 & 72 \end{bmatrix} \begin{bmatrix} R_C \\ C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 550 \\ 67.5 \\ 4567.5 \end{bmatrix} \quad (13)$$

Solving this using numerical software like Python gives:

$$R_C = 70.8, \quad C_1 = 12.14, \quad C_2 = -20.54 \quad (14)$$

The reactions at A are obtained by setting $x = 0$ for $V(x)$ and $M(x)$:

$$R_A = V(0) = C_1 = 12.14 \quad (15)$$

$$M_A = -M(0) = -C_2 = 20.54 \quad (16)$$

Note that the anti-clockwise M_A causes a hogging moment. The reaction at D is obtained by setting $x = 12$, which gives:

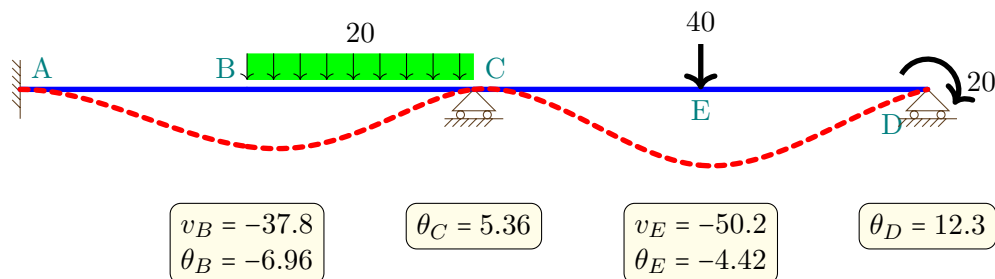
$$R_D = -V(12) = -(-20 \cdot 9 + 20 \cdot 6 + 70.8 - 40 + 12.14) = 17.06 \quad (17)$$

We can verify that the equilibrium equations are satisfied:

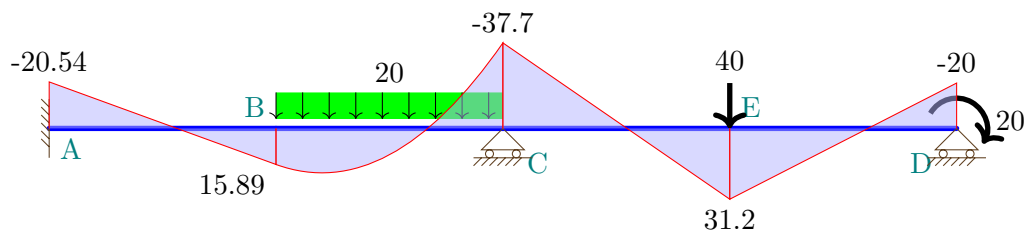
$$\sum F_y = R_A + R_C + R_D - 20 \cdot 3 - 40 = 12.14 + 70.8 + 17.06 - 60 - 40 = 0 \quad (18)$$

$$\begin{aligned} \sum M_{\odot A} &= M_A + R_C \cdot 6 + R_D \cdot 12 - 20 \cdot 3 \cdot 4.5 - 40 \cdot 9 - 20 \\ &= 20.54 + 70.8 \cdot 6 + 17.06 \cdot 12 - 20 \cdot 3 \cdot 4.5 - 40 \cdot 9 - 20 \\ &= 0.06 \approx 0 \end{aligned} \quad (19)$$

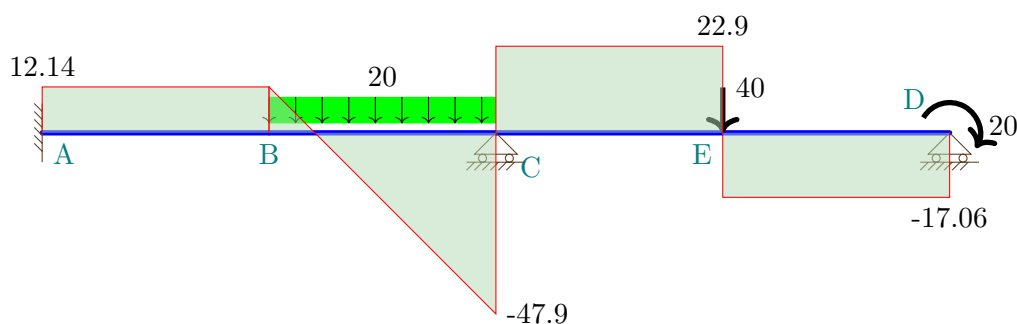
Deflected Shape (values are normalised by $1/EI$):



BMD:

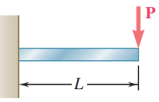
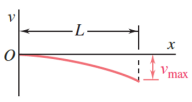
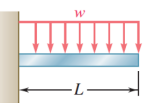
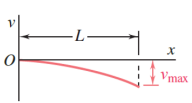
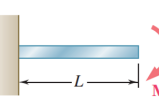
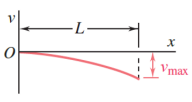


SFD:



Question 2 Superposition Method for Beam Analysis

For the same beam as in the previous question, use the superposition method and the following table to solve for the reactions, the rotation (i.e., slope) at D, and the deflection at B.

Beam and Loading	Elastic Curve	Maximum Deflection	Slope at End	Equation of Elastic Curve
1 		$-\frac{PL^3}{3EI}$	$-\frac{PL^2}{2EI}$	$v = \frac{P}{6EI}(x^3 - 3Lx^2)$
2 		$-\frac{wL^4}{8EI}$	$-\frac{wL^3}{6EI}$	$v = -\frac{w}{24EI}(x^4 - 4Lx^3 + 6L^2x^2)$
3 		$-\frac{M_bL^2}{2EI}$	$-\frac{M_bL}{EI}$	$v = -\frac{M_b}{2EI}x^2$

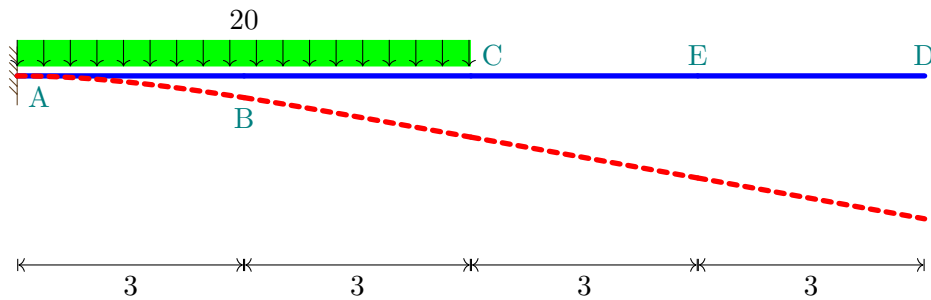
Analysis Guide:

- The beam type provided in the given table is a cantilever. So, you will need to create such a beam in the solution process.
- An unloaded beam segment with a free end has no bending moment or shear. Therefore, it remains straight, undergoing rigid body motion following the displacement and rotation at the supported end.
- A beam with a partially distributed load would require the separation of a few load cases.
- Answers for checking: $v_B = -37.8/EI$, $\theta_D = 12.3/EI$.

Solution:

The load and the reactions at C and D can be split into the following cases:

load 1



$$v_{B1} = -\frac{20}{24EI}(3^4 - 4 \cdot 6 \cdot 3^3 + 6 \cdot 6^2 \cdot 3^2) = -1147.5/EI \quad (20)$$

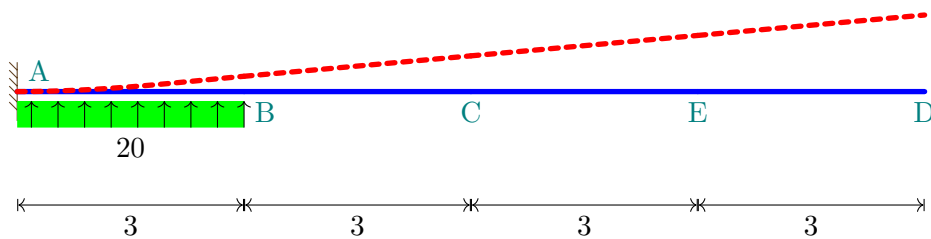
$$v_{C1} = -\frac{20 \cdot 6^4}{8EI} = -3240/EI \quad (21)$$

$$\theta_{C1} = -\frac{20 \cdot 6^3}{6EI} = -720/EI \quad (22)$$

$$v_{D1} = v_{C1} + \theta_{C1} \cdot 6 = -7560/EI \quad (23)$$

$$\theta_{D1} = \theta_{C1} = -720/EI \quad (24)$$

load 2



$$v_{B2} = \frac{20 \cdot 3^4}{8EI} = 202.5/EI \quad (25)$$

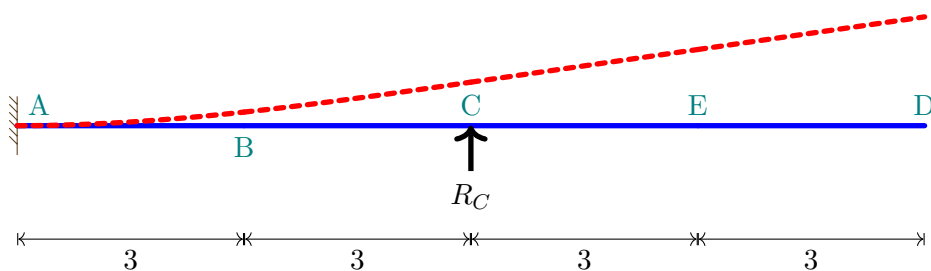
$$\theta_{B2} = \frac{20 \cdot 3^3}{6EI} = 90/EI \quad (26)$$

$$v_{C2} = v_{B2} + \theta_{B2} \cdot 3 = 472.5/EI \quad (27)$$

$$v_{D2} = v_{B2} + \theta_{B2} \cdot 9 = 1012.5/EI \quad (28)$$

$$\theta_{D2} = \theta_{B2} = 90/EI \quad (29)$$

load 3



$$v_{B3} = \frac{-R_C}{6EI} (3^3 - 3 \cdot 6 \cdot 3^2) = 22.5R_C/EI \quad (30)$$

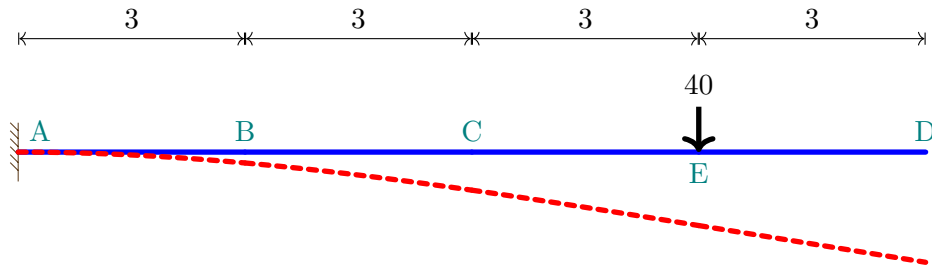
$$v_{C3} = \frac{R_C \cdot 6^3}{3EI} = 72R_C/EI \quad (31)$$

$$\theta_{C3} = \frac{R_C \cdot 6^2}{2EI} = 18R_C/EI \quad (32)$$

$$v_{D3} = v_{C3} + \theta_{C3} \cdot 6 = 180R_C/EI \quad (33)$$

$$\theta_{D3} = \theta_{C3} = 18R_C/EI \quad (34)$$

load 4



$$v_{B4} = \frac{40}{6EI} (3^3 - 3 \cdot 9 \cdot 3^2) = -1440/EI \quad (35)$$

$$v_{C4} = \frac{40}{6EI} (6^3 - 3 \cdot 9 \cdot 6^2) = -5040/EI \quad (36)$$

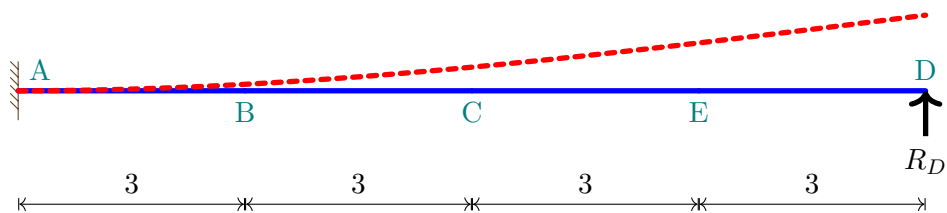
$$v_{E4} = \frac{-40 \cdot 9^3}{3EI} = -9720/EI \quad (37)$$

$$\theta_{E4} = \frac{-40 \cdot 9^2}{2EI} = -1620/EI \quad (38)$$

$$v_{D4} = v_{E4} + \theta_{E4} \cdot 3 = -14580/EI \quad (39)$$

$$\theta_{D4} = \theta_{E4} = -1620/EI \quad (40)$$

load 5



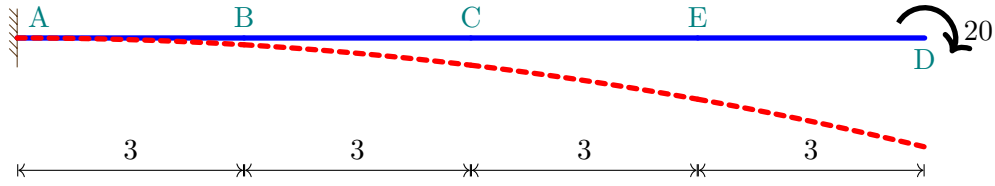
$$v_{B5} = \frac{-R_D}{6EI} (3^3 - 3 \cdot 12 \cdot 3^2) = 49.5R_D/EI \quad (41)$$

$$v_{C5} = \frac{-R_D}{6EI} (6^3 - 3 \cdot 12 \cdot 6^2) = 180R_D/EI \quad (42)$$

$$v_{D5} = \frac{R_D \cdot 12^3}{3EI} = 576R_D/EI \quad (43)$$

$$\theta_{D5} = \frac{R_D \cdot 12^2}{2EI} = 72R_D/EI \quad (44)$$

load 6



$$v_{B6} = \frac{-20}{2EI}(3^2) = -90/EI \quad (45)$$

$$v_{C6} = \frac{-20}{2EI}(6^2) = -360/EI \quad (46)$$

$$v_{D6} = \frac{-20}{2EI}(12^2) = -1440/EI \quad (47)$$

$$\theta_{D6} = \frac{-20}{EI}(12) = -240/EI \quad (48)$$

The compatibility conditions require $v_C = 0$ and $v_D = 0$. Therefore

$$0 = v_C = v_{C1} + v_{C2} + v_{C3} + v_{C4} + v_{C5} + v_{C6} \quad (49)$$

$$\implies 0 = -3240 + 472.5 + 72R_C - 5040 + 180R_D - 360 \quad (50)$$

$$0 = v_D = v_{D1} + v_{D2} + v_{D3} + v_{D4} + v_{D5} + v_{D6} \quad (51)$$

$$\implies 0 = -7560 + 1012.5 + 180R_C - 14580 + 576R_D - 1440 \quad (52)$$

Solving the two equations above gives $R_C = 70.8$ and $R_D = 17.06$. Afterward, we can obtain the deflection at B and the rotation at D:

$$v_B = v_{B1} + v_{B2} + v_{B3} + v_{B4} + v_{B5} + v_{B6} \quad (53)$$

$$= (-1147.5 + 202.5 + 22.5R_C - 1440 + 49.5R_D - 90)/EI \quad (54)$$

$$= -37.8/EI \quad (55)$$

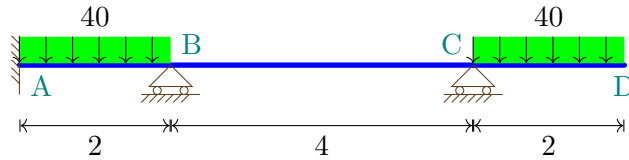
$$\theta_D = \theta_{D1} + \theta_{D2} + \theta_{D3} + \theta_{D4} + \theta_{D5} + \theta_{D6} \quad (56)$$

$$= (-720 + 90 + 18R_C - 1620 + 72R_D - 240)/EI \quad (57)$$

$$= 12.3/EI \quad (58)$$

Question 3 Moment-Area Theorems for Beam Analysis

Consider the following beam with a uniform flexural rigidity EI . Use the moment-area theorems to solve for its reactions and draw the BMD, SFD, and deflected shape.



Analysis Guide:

- Be careful with how the redundant reactions are selected. Some choices will result in a very complex BMD shape that requires many subdivisions into simpler shapes with well-known formulas for area and centroid. Other choices will result in a much simpler BMD shape, making it easier to compute their areas and centroids.
- If you choose to use the reactions at B and C as the redundants, consider splitting the two UDLs into two separate load cases. Otherwise, the BMD within span AB would need to be split into multiple shapes before a parabolic spandrel can be obtained.
- Answers for checking: $R_A = 69.1$, $M_A = 32.7$.

Solution:

The degree of static indeterminacy (SI) of the beam is 2. Therefore, we need two redundants. We also need to identify two compatibility conditions to solve for these two redundants.

The displacement boundary conditions of the beam are:

$$v_A = 0, \quad \theta_A = 0, \quad v_B = 0, \quad v_C = 0 \quad (59)$$

We can use the 2nd moment-area theorem to relate all these displacement conditions:

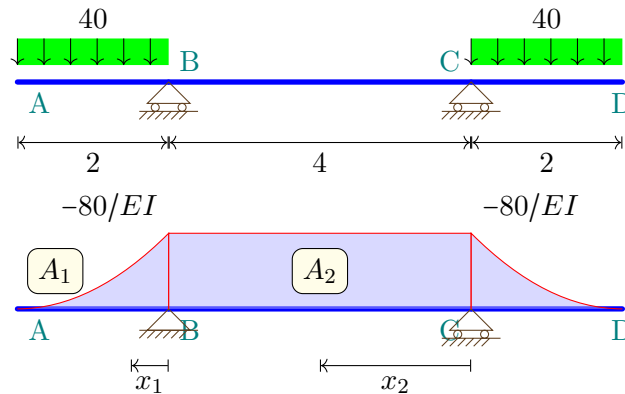
$$v_B = v_A + \theta_A L_{AB} + \delta_{BA} \implies \delta_{BA} = 0 \quad (60)$$

$$v_C = v_A + \theta_A L_{AC} + \delta_{CA} \implies \delta_{CA} = 0 \quad (61)$$

These are the two compatibility conditions that can help us determine the two unknown redundants. Note that δ_{BA} is the moment of the curvature area between A and B taken about B, while δ_{CA} is the moment of the curvature area between A and C taken about C.

Let the moment M_A and the vertical reaction R_A at A be the two redundants. With these two redundants, there are three load cases and their corresponding curvature diagrams.

load 1: with the redundants removed



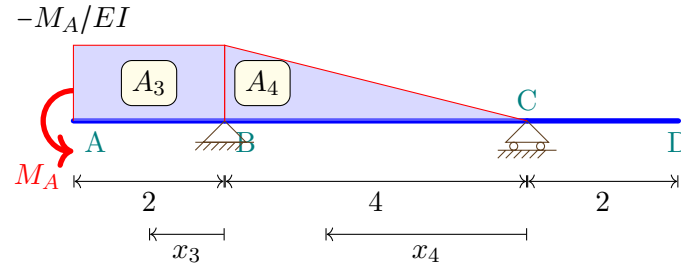
$$A_1 = -80 \cdot 2 / (3EI) = -160 / (3EI) \quad (62)$$

$$x_1 = 2/4 = 1/2 \quad (63)$$

$$A_2 = -80 \cdot 4 / EI = -320 / EI \quad (64)$$

$$x_2 = 4/2 = 2 \quad (65)$$

load 2: with M_A only



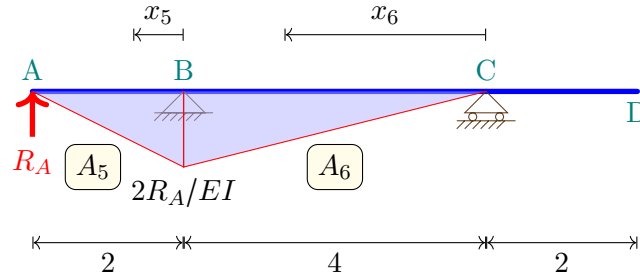
$$A_3 = -M_A/EI \cdot 2 = -2M_A/EI \quad (66)$$

$$x_3 = 2/2 = 1 \quad (67)$$

$$A_4 = -M_A/EI \cdot 4/2 = -2M_A/EI \quad (68)$$

$$x_4 = 4 \cdot 2/3 = 8/3 \quad (69)$$

load 3: with R_A only



$$A_5 = 2R_A/EI \cdot 2/2 = 2R_A/EI \quad (70)$$

$$x_5 = 2/3 \quad (71)$$

$$A_6 = 2R_A/EI \cdot 4/2 = 4R_A/EI \quad (72)$$

$$x_6 = 4 \cdot 2/3 = 8/3 \quad (73)$$

Substituting the above into the compatibility conditions gives:

$$0 = \delta_{BA} = A_1x_1 + A_3x_3 + A_5x_5 \quad (74)$$

$$\implies 0 = -160/3 \cdot 1/2 - 2M_A \cdot 1 + 2R_A \cdot 2/3 \quad (75)$$

$$\implies 80 = -6M_A + 4R_A \quad (76)$$

$$0 = \delta_{CA} = A_1(4 + x_1) + A_2x_2 + A_3(4 + x_3) + A_4x_4 + A_5(4 + x_5) + A_6x_6 \quad (77)$$

$$\implies 0 = -160/3 \cdot (4 + 1/2) - 320 \cdot 2 - 2M_A \cdot 8/3 - 2M_A \cdot (4 + 1) + 2R_A \cdot (4 + 2/3) + 4R_A \cdot 8/3 \quad (78)$$

$$\implies 2640 = -46M_A + 60R_A \quad (79)$$

Solving the above two equations gives $M_A = 32.7$ and $R_A = 69.1$.

Using the global equilibrium equations to obtain the remaining reactions:

$$\sum F_y = 0 \implies 40 \cdot (2 + 2) = R_A + R_B + R_C \quad (80)$$

$$\implies R_B + R_C = 160 - 69.1 = 90.9 \quad (81)$$

$$\sum M_{\oplus A} = 0 \implies 0 = M_A + 2R_B + 6R_C - 40 \cdot 4 \cdot 4 \quad (82)$$

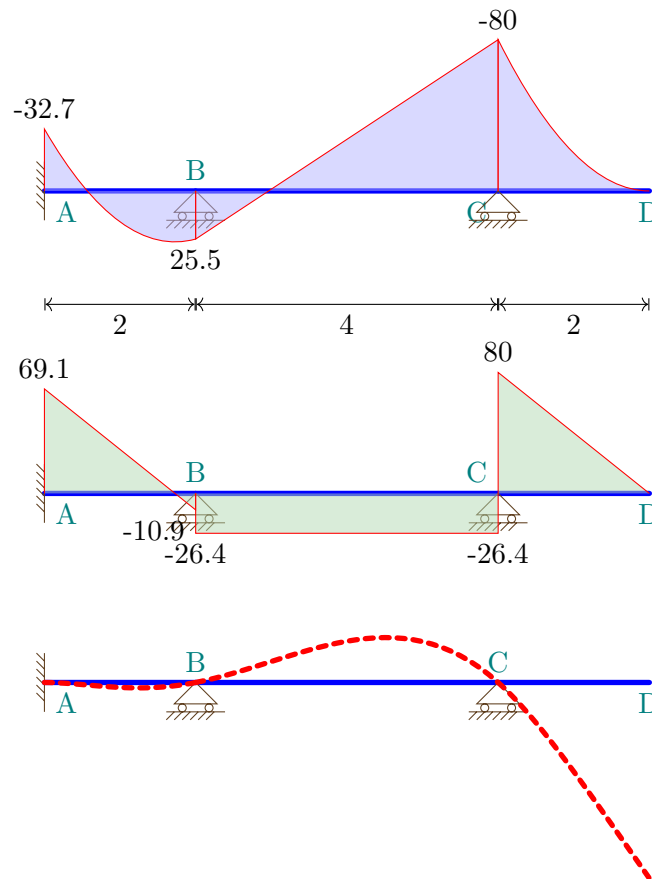
$$\implies R_B + 3R_C = 320 - 32.7/2 = 303.64 \quad (83)$$

Solving the above gives $R_B = -15.5$ and $R_C = 106.4$.

Let's compute the bending moment at B, i.e., M_b (note that M_B is the notation for an external moment applied at B, not the internal bending moment at B).

$$M_b = -80 - M_A + 2R_A = 25.5 \quad (84)$$

Let's plot the bending moment and shear force diagrams, as well as the deflected shape:



Note,

$$\theta_B = A_1 + A_3 + A_5 = 19.4/EI \quad (85)$$

$$\theta_C = \sum_{i=1}^6 A_i = -89.7/EI \quad (86)$$

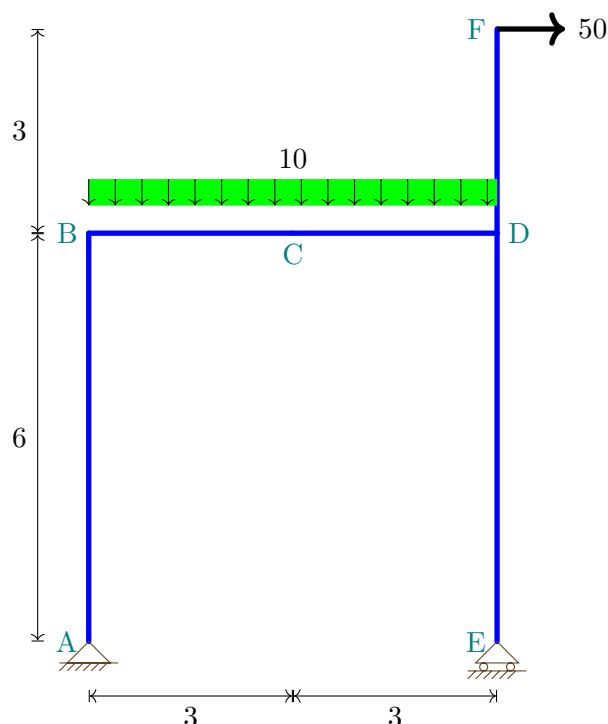
$$\theta_D = \theta_C - 80 \cdot 2/(3EI) = -143/EI \quad (87)$$

$$v_D = v_C + \theta_C \cdot L_{CD} + \delta_{DC} \quad (88)$$

$$= -89.7/EI \cdot 2 - 80 \cdot 2/(3EI) \cdot (2 \cdot 3/4) = -259.4/EI \quad (89)$$

Question 4 Statically Determinate Frame

Consider the following frame. Use the equilibrium equations to solve for its reactions and draw the BMD, SFD, and deflected shape.



Analysis Guide:

- First, calculate the support reactions using the equilibrium equations, i.e., $\sum F_X = 0$, $\sum F_Y = 0$, and $\sum M_Z = 0$.
- Draw the free-body diagram for each member and determine the unknown forces and moments p and m acting at the ends of each member using equilibrium equations.
- When transferring the p and m of each member to the connected joints, remember to reverse the directions of the forces and moments (action and reaction).
- Ensure the forces and moments at each joint satisfy the equilibrium equations.
- When drawing the BMD and SFD for the entire structure, it is advisable to start with the BMD and SFD for each individual member. Label the values of all critical points, such as local maxima and minima.
- When sketching the deflected shape, remember that members without a bending moment remain straight.
- The rotation at point D may not be immediately clear due to the competing effects of the UDL and the point load at F. To begin, assume the rotation at point D is positive, i.e., $\theta_D > 0$.
- Answers for checking: $R_{Ax} = -50$, $R_{Ay} = -45$, $R_{Ey} = 105$.

Solution:

First, use the global equilibrium equations to calculate the three reaction forces:

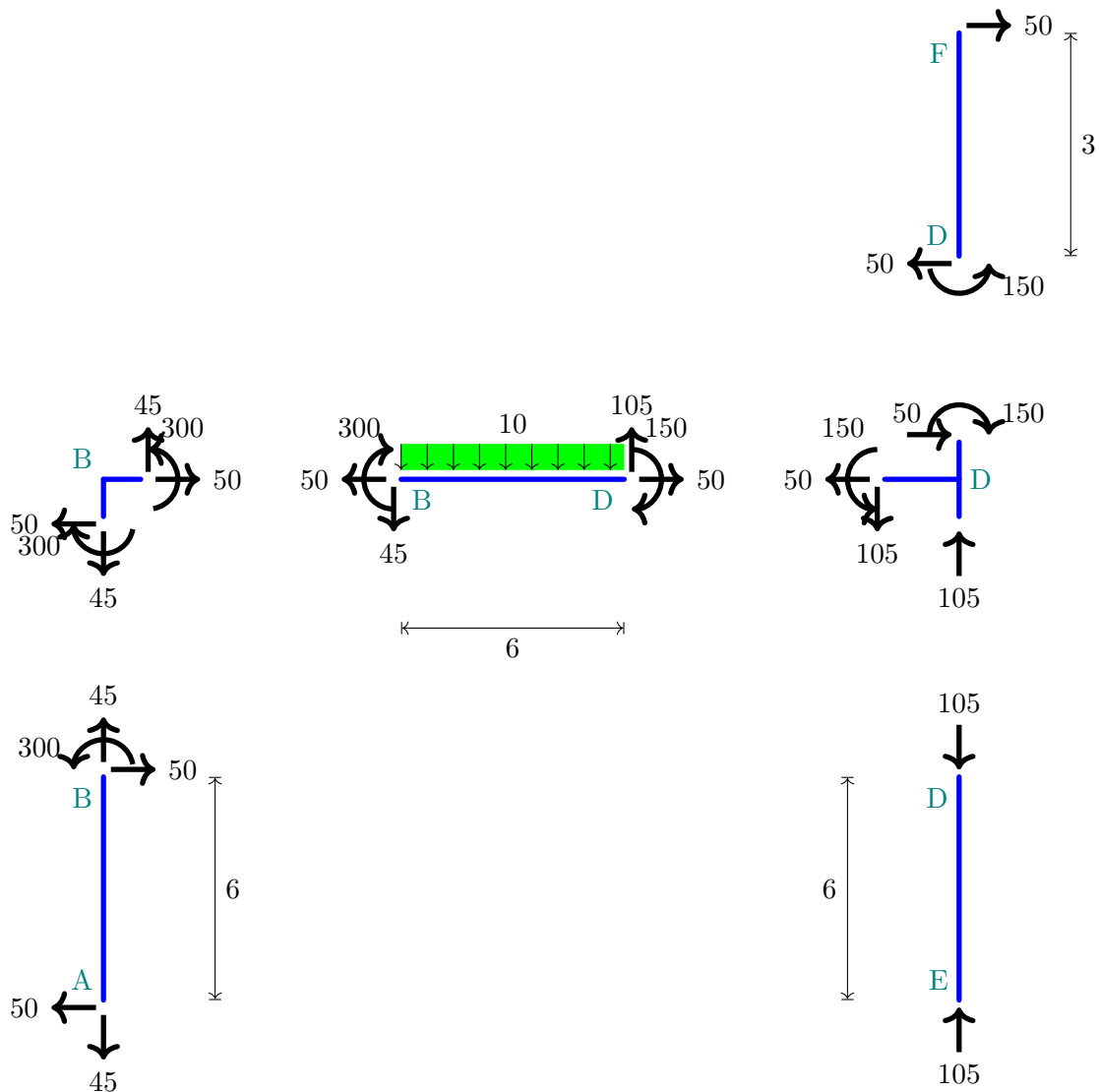
$$\sum M_{\odot A} = 0 = R_{Ey} \cdot 6 - 50 \cdot 9 - 10 \cdot 6 \cdot 3 \quad (90)$$

$$\implies R_{Ey} = 105 \quad (91)$$

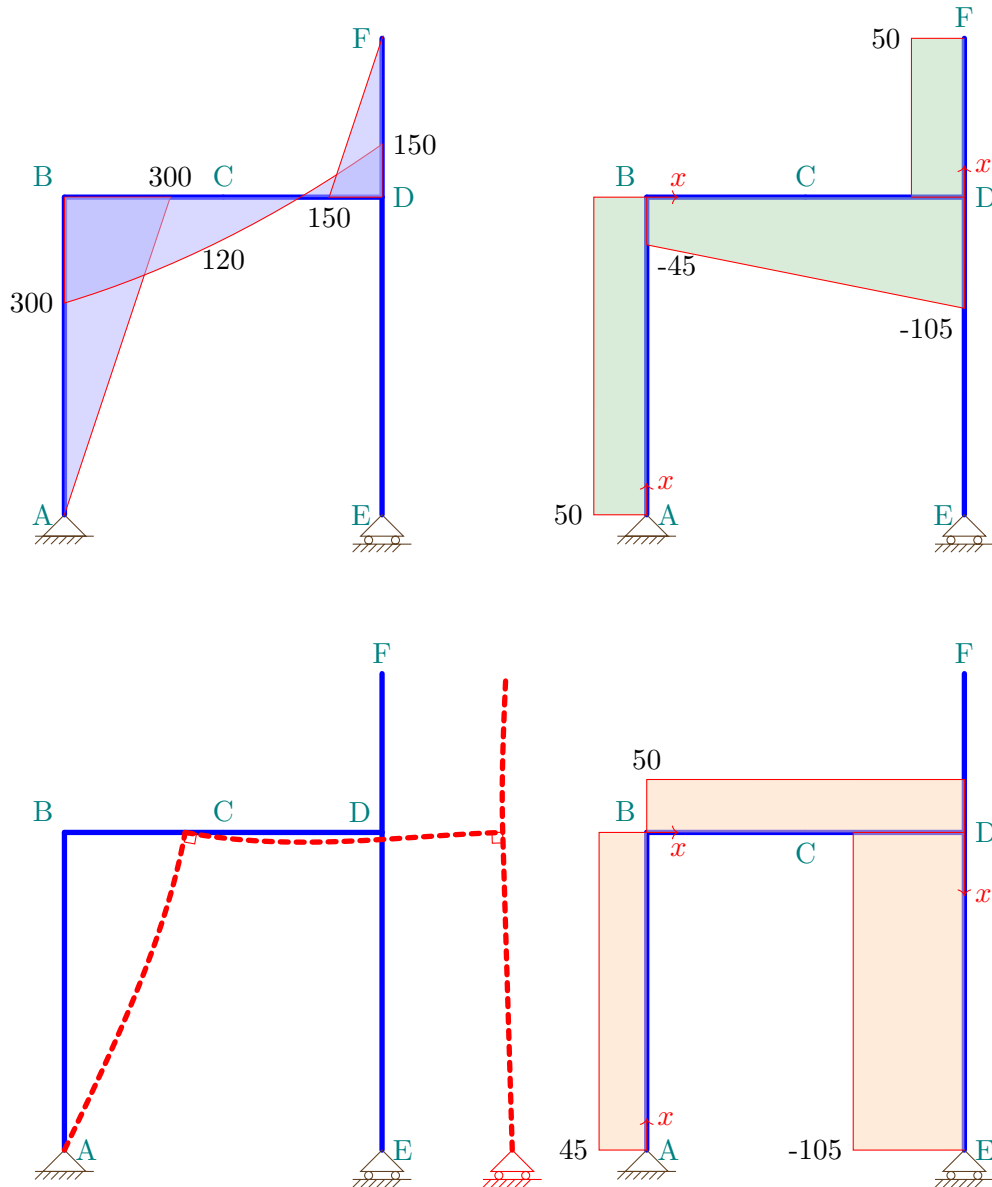
$$\sum F_x = 0 = R_{Ax} + 50 \implies R_{Ax} = -50 \quad (92)$$

$$\sum F_y = 0 = R_{Ay} + R_{Ey} - 10 \cdot 6 \implies R_{Ay} = -45 \quad (93)$$

Next, draw the free-body diagrams of all members and joints, and calculate the forces and moments at the member ends and joints using the equilibrium equations:



Start the calculations for members AB, DE, and DF (using the known reactions), then transfer the forces and moments to joints B and D and to member BD. Check the forces on member BD against the equilibrium equations for correctness.

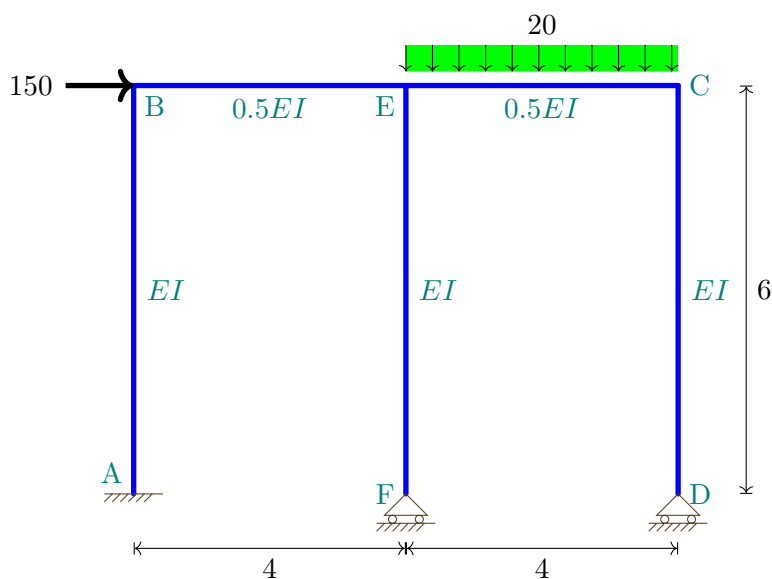


Note that in the deflected shape, member DE remains straight (as there is no bending moment), while member DF becomes curved.

Question 5 Moment-Area Theorems and Unit-Load Method for Frame Analysis

Consider the following frame with different flexural rigidity for different members.

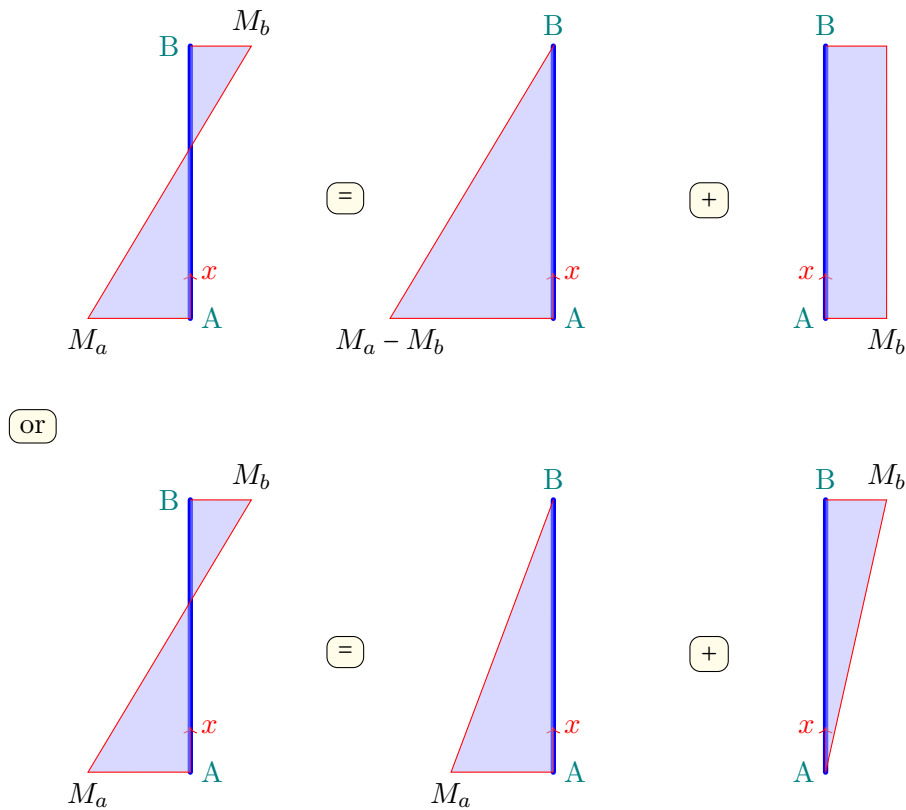
- Use the **moment-area theorems** to solve for the support reactions, and draw the BMD, SFD, AFD, and deflected shape. Calculate the horizontal deflection at C and rotation at F.
- Suppose the roller support at D is accidentally removed**, use the **Unit-Load method** to analyse the frame and calculate the horizontal deflection at B.



Analysis Guide:

- Be careful when selecting the redundant reactions. Not all options will result in a BMD that is easy to compute for areas and centroids.
- When dividing the BMD into simpler shapes, label each shape with its area A_i and centroid x_i to facilitate checking.
- Decide on the orientation of each column's x -axis, as well as the value of its s . It is common to follow the alphabetical order of the joint labels unless stated otherwise.
- The beam's x -axis is commonly chosen to start from the left end unless stated otherwise.
- When converting the BMD into the curvature diagram, remember that the flexural rigidity for members BE and EC is $0.5EI$.
- Be careful to use the correct sign for each A_i corresponding to the orientation of each member's x -axis.
- When using the second moment-area theorem for each column, remember to include s to ensure the correct sign of the resultant displacement.

- In frame analysis, the axial deformations of members are typically ignored because they are several orders of magnitude smaller than the flexural deformations.
- If necessary, a BMD shape with double curvature bending can be represented as the sum of two simpler shapes, as shown below with two options, where M_a and M_b are signed values. This separation avoids the need to find the point where $M = 0$.



- Answers for checking: (a) With the support at point D: $u_C = 4910.4/EI$, $\theta_F = 164.8/EI$; (b) Without the support at point D: $u_B = 4749.3/EI$.

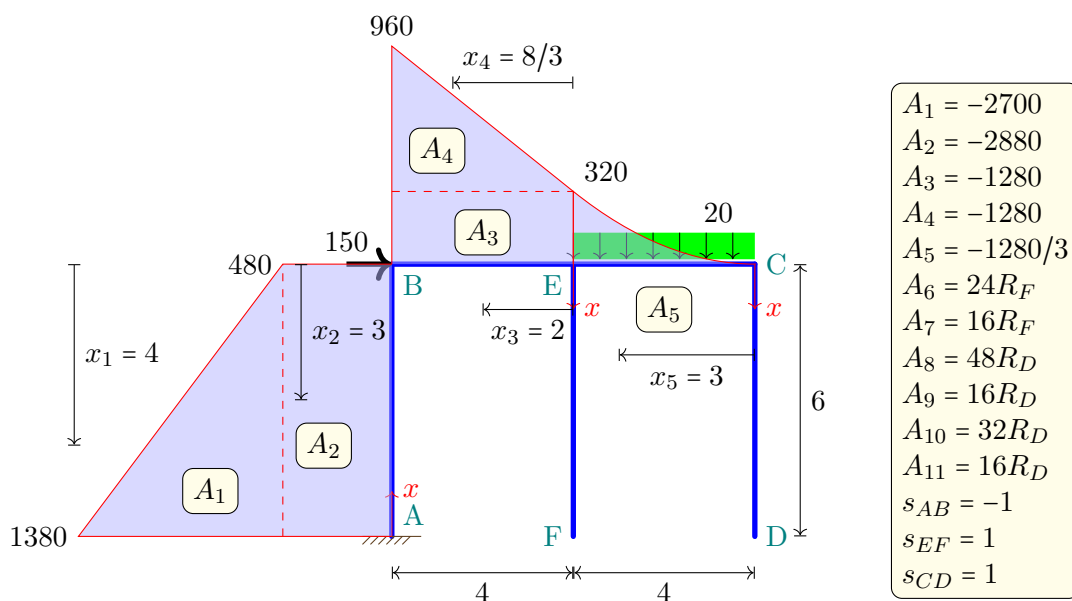
Solution:

Part (a)

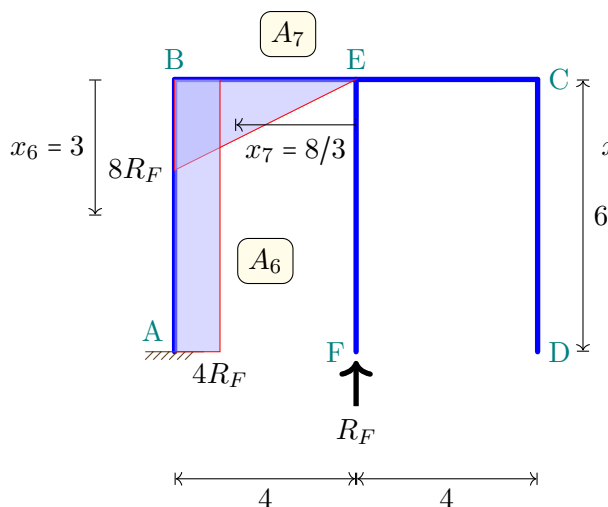
Select the vertical reactions at D and F as the two redundants, with $v_D = 0$ and $v_F = 0$ as the corresponding compatibility conditions. However, since members CD and EF are assumed to be axially rigid, $v_C = v_D$ and $v_E = v_F$. Therefore, we can use $v_C = 0$ and $v_E = 0$ as the compatibility conditions.

We can create three loading cases for the applied loads and the redundants, and plot their corresponding curvature diagrams as follows. **Note: EI can be treated as the unit for flexural rigidity and is omitted from the intermediate calculations for simplicity.**

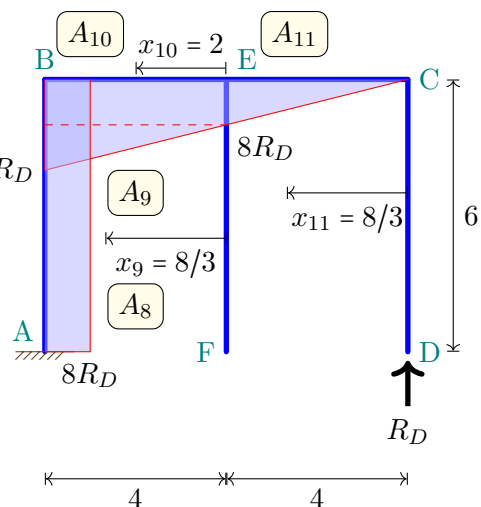
Case 1: Redundants removed



Case 2: R_F



Case 3: R_D



Case 1:

$$\theta_{B1} = A_1 + A_2 = -5580 \quad (94)$$

$$\theta_{E1} = \theta_{B1} + A_3 + A_4 = -8140 \quad (95)$$

$$\theta_{C1} = \theta_{E1} + A_5 = -25700/3 \quad (96)$$

$$u_{B1} = s_{AB} (A_1 \cdot x_1 + A_2 \cdot x_2) = 19440 \quad (97)$$

$$v_{E1} = \theta_{B1} \cdot L_{BE} + \delta_{BE1} (= A_3 \cdot x_3 + A_4 \cdot x_4) = -84880/3 \quad (98)$$

$$v_{C1} = v_{E1} + \theta_{E1} \cdot L_{EC} + \delta_{EC1} (= A_5 \cdot x_5) = -186400/3 \quad (99)$$

Case 2:

$$\theta_{B2} = A_6 = 24R_F \quad (100)$$

$$\theta_{E2} = \theta_{B2} + A_7 = 40R_F \quad (101)$$

$$\theta_{C2} = \theta_{E2} = 40R_F \quad (102)$$

$$u_{B2} = s_{AB} (A_6 \cdot x_6) = -72R_F \quad (103)$$

$$v_{E2} = \theta_{B2} \cdot L_{BE} + \delta_{BE2} (= A_7 \cdot x_7) = 416R_F/3 \quad (104)$$

$$v_{C2} = v_{E2} + \theta_{E2} \cdot L_{EC} = 896R_F/3 \quad (105)$$

Case 3:

$$\theta_{B3} = A_8 = 48R_D \quad (106)$$

$$\theta_{E3} = \theta_{B3} + A_9 + A_{10} = 96R_D \quad (107)$$

$$\theta_{C3} = \theta_{E3} + A_{11} = 112R_D \quad (108)$$

$$u_{B3} = s_{AB} (A_8 \cdot x_8) = -144R_D \quad (109)$$

$$v_{E3} = \theta_{B3} \cdot L_{BE} + \delta_{BE3} (= A_9 \cdot x_9 + A_{10} \cdot x_{10}) = 896R_D/3 \quad (110)$$

$$v_{C3} = v_{E3} + \theta_{E3} \cdot L_{EC} + \delta_{EC3} (= A_{11} \cdot x_{11}) = 2176R_D/3 \quad (111)$$

Enforce the compatibility conditions:

$$v_E = v_{E1} + v_{E2} + v_{E3} = -84880/3 + 416R_F/3 + 896R_D/3 = 0 \quad (112)$$

$$v_C = v_{C1} + v_{C2} + v_{C3} = -186400/3 + 896R_F/3 + 2176R_D/3 = 0 \quad (113)$$

Solving the above equations yields $R_F = 172.7$ and $R_D = 14.55$. Therefore, the vertical reaction and moment at A are $R_{Ay} = 20 \times 4 - R_F - R_D = -107.25$ and $R_{Ax} = -150$.

We can combine the results of all three cases to obtain the following (with EI included):

$$u_B = u_E = u_C = 4910.4/EI \quad (114)$$

$$\theta_B = -736.8/EI \quad (115)$$

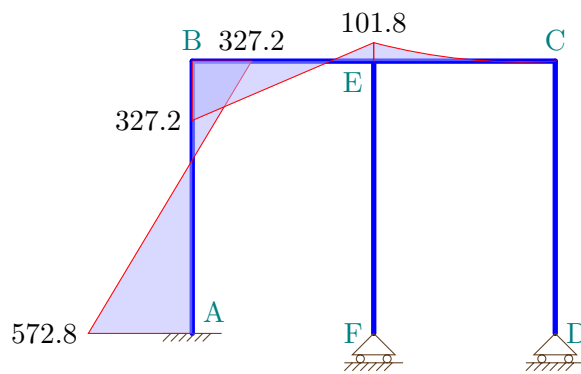
$$\theta_C = \theta_D = -29.1/EI \quad (116)$$

$$\theta_E = \theta_F = 164.8/EI \quad (117)$$

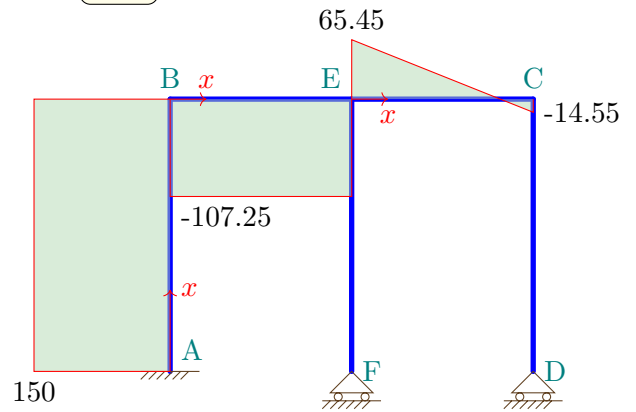
$$u_F = u_E + s_{EF} (\theta_E \cdot L_{EF}) = 5899.2/EI \quad (118)$$

$$u_D = u_C + s_{CD} (\theta_C \cdot L_{CD}) = 4736.2/EI \quad (119)$$

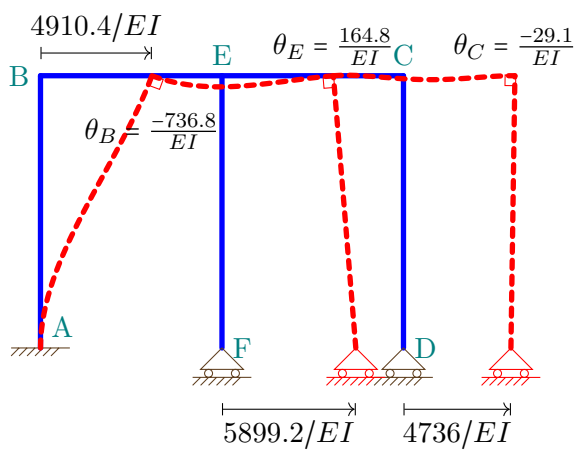
BMD



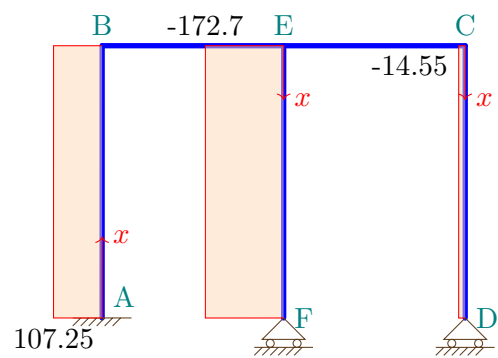
SFD



Deflected Shape

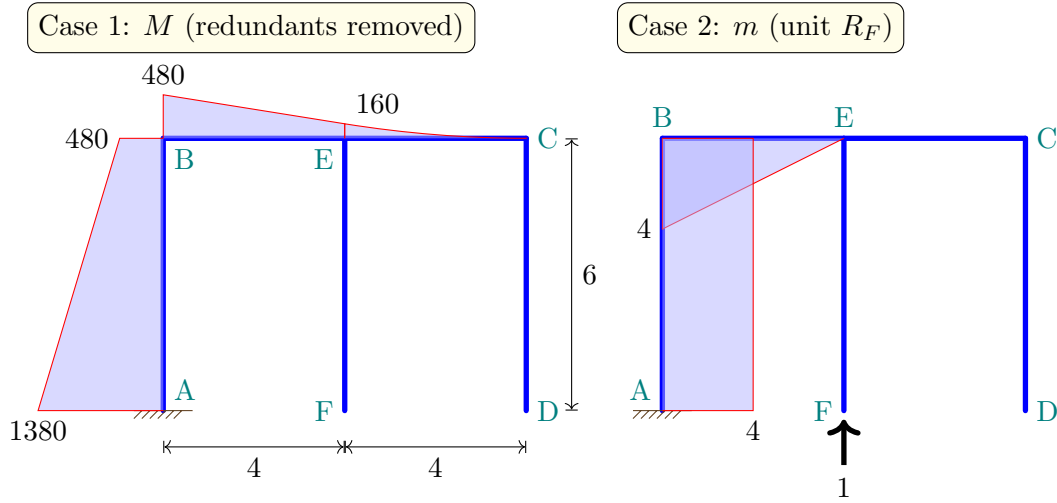


AFD



Part (b)

When the support at point D is removed, the bending moment diagrams for Cases 1 and 2 are as follows:



Compute $\int \frac{mM}{EI} dx$ and $\int \frac{m^2}{EI} dx$ (note that the flexural rigidity of member BE is $0.5EI$).

Based on the above diagrams, computation is only required for members AB and BE.

$$\int \frac{mM}{EI} dx = \frac{1}{EI} \times \frac{1}{2} \times 4 \times (-1380 - 480) \times 6 \quad (\text{member AB}) \quad (120)$$

$$+ \frac{1}{0.5EI} \times \frac{1}{6} \times 4 \times (-160 - 2 \times 480) \times 4 \quad (\text{member BE}) \quad (121)$$

$$= -\frac{84880}{3EI} \quad (122)$$

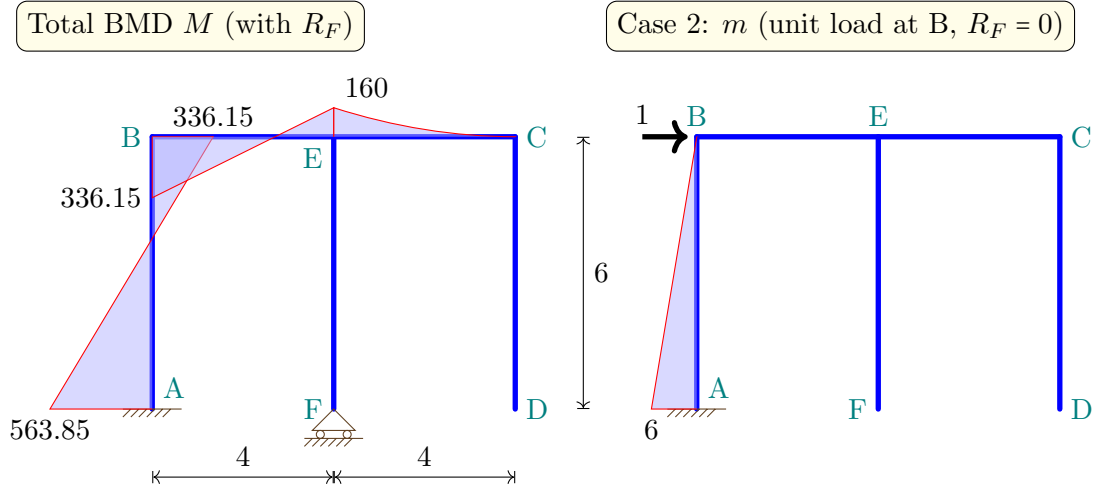
$$\int \frac{m^2}{EI} dx = \frac{1}{EI} \times 4 \times 4 \times 6 \quad (\text{member AB}) \quad (123)$$

$$+ \frac{1}{0.5EI} \times \frac{1}{3} \times 4 \times 4 \times 4 \quad (\text{member BE}) \quad (124)$$

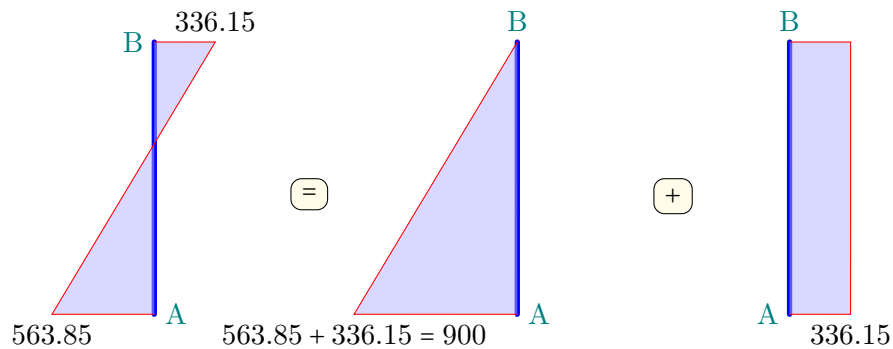
$$= \frac{416}{3EI} \quad (125)$$

$$\Rightarrow R_F = \frac{-\int \frac{mM}{EI} dx}{\int \frac{m^2}{EI} dx} = 84880/416 = 204.04 \quad (126)$$

To compute the horizontal displacement at point B, let's draw the BMDs for the total response (including R_F) and for a unit horizontal load applied at point B:



Based on the above diagrams, computation is only required for member AB. However, the moment integration for the BMD M of member AB is not provided in the table. We can express the BMD M of member AB as the sum of a hogging (negative) triangular M and a sagging (positive) rectangular M , as shown below:



Therefore,

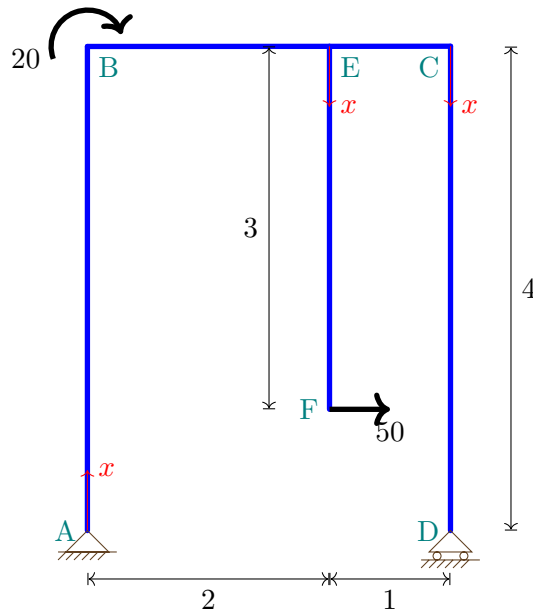
$$u_B = \sum \int \frac{mM}{EI} dx = \frac{1}{EI} \times \frac{1}{3} \times (-6) \times (-900) \times 6 \quad (\text{hogging triangular } M) \quad (127)$$

$$+ \frac{1}{EI} \times \frac{1}{2} \times (-6) \times 336.15 \times 6 \quad (\text{sagging rectangular } M) \quad (128)$$

$$= 4749.3/EI \quad (129)$$

Question 6 Unit-Load Method for Frame Analysis

Consider the following frame with uniform flexural rigidity EI . Use the global equilibrium equations to calculate the support reactions, and draw the BMD, SFD, and AFD. Sketch the qualitative deflected shape, and use the unit-load method to compute the horizontal deflection at point D and the rotation at point C.



Analysis Guide:

- The orientation of each column's x -axis has been provided.
- Be careful of the sign of M and m in each member.
- Answers for checking: $u_D = 8060/(3EI)$, $\theta_C = 135/EI$.

Solution:

The support reactions can be found by using the global equilibrium equations:

$$R_D = \frac{20 + 50 \times 1}{3} = 70/3 \quad (130)$$

$$R_{Ax} = -50 \quad (131)$$

$$R_{Ay} = -70/3 \quad (132)$$

The BMD, SFD, AFD, and qualitative deflected shape are shown on the next page.

The calculations for u_D and θ_C using the unit-load method are given as follows, with the BMD m of these two cases given on the next page.

$$u_D = \sum \int \frac{mM}{EI} dx = \frac{1}{EI} \times \frac{1}{3} \times 4 \times 200 \times 4 \quad (\text{member AB}) \quad (133)$$

$$+ \frac{1}{EI} \times \frac{1}{2} \times 4 \times (220 + 520/3) \times 2 \quad (\text{member BE}) \quad (134)$$

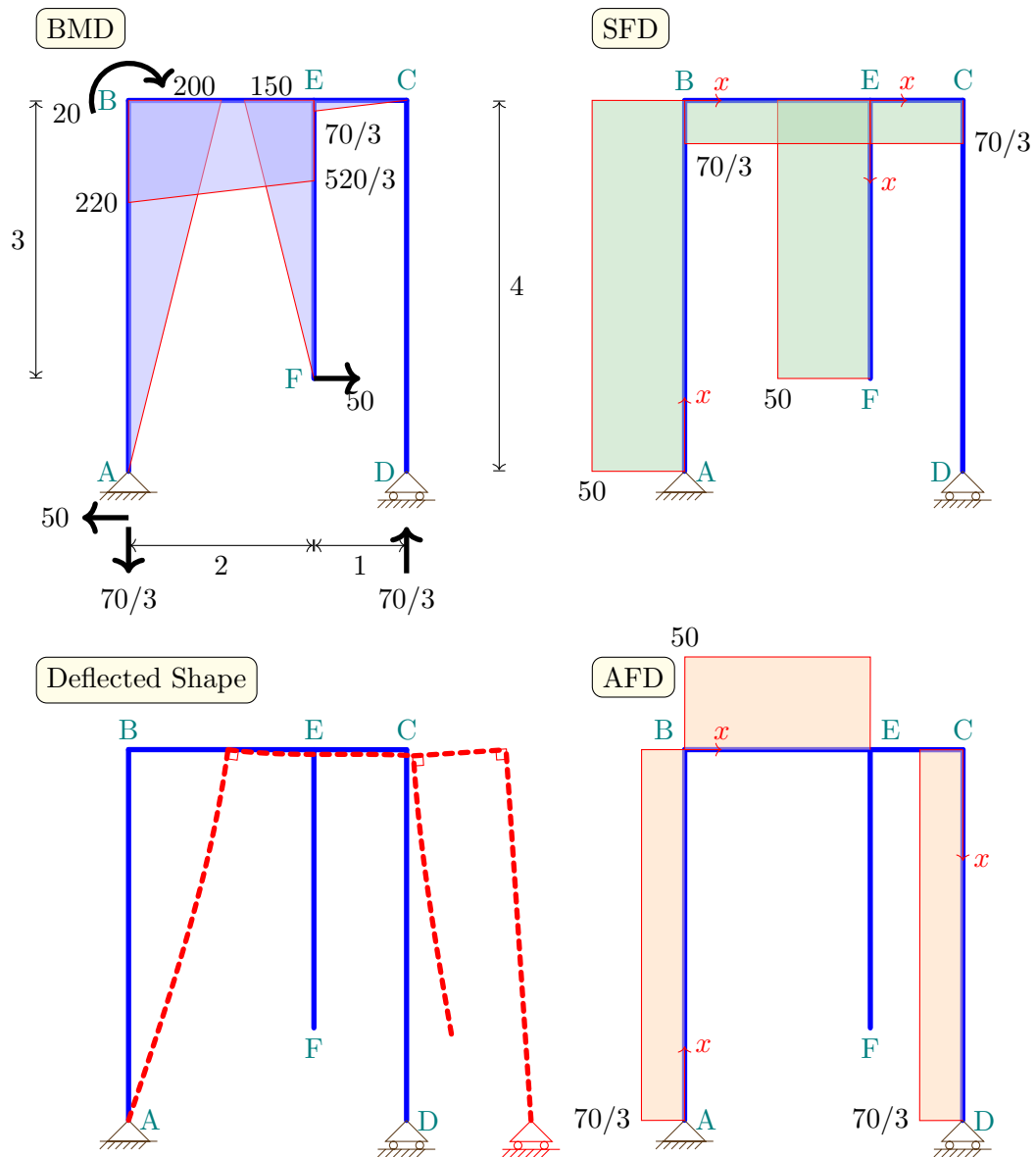
$$+ \frac{1}{EI} \times \frac{1}{2} \times 4 \times 70/3 \times 1 \quad (\text{member EC}) \quad (135)$$

$$= 8060/(3EI) \quad (136)$$

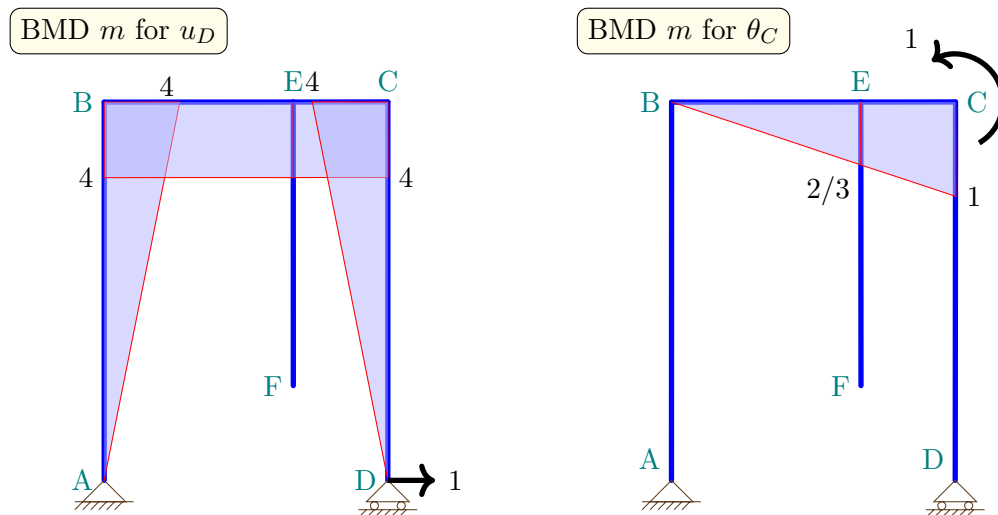
$$\theta_C = \sum \int \frac{mM}{EI} dx = \frac{1}{EI} \times \frac{1}{6} \times \frac{2}{3} \times (220 + 2 \times 520/3) \times 2 \quad (\text{member BE}) \quad (137)$$

$$+ \frac{1}{EI} \times \frac{1}{6} \times 70/3 \times (1 + 2 \times 2/3) \times 1 \quad (\text{member EC}) \quad (138)$$

$$= 135/EI \quad (139)$$



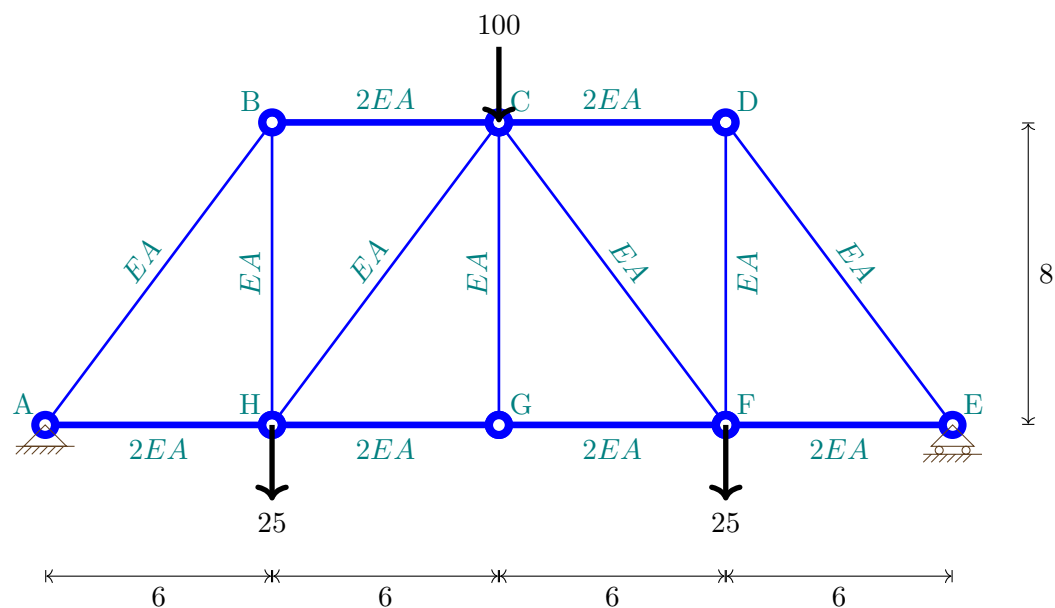
To compute the deflection at D and rotation at C by applying one horizontal unit-load at D and one unit-moment at C, with their corresponding BMD shown as follows



Question 7 Unit-Load Method for Truss Analysis

Consider the following truss, where the axial rigidity is $2EA$ for the members in the top and bottom chords and EA for the diagonal and vertical members.

- Calculate the vertical deflection at point G and the horizontal displacement at point E.
- If the support at point E were a pin (not a roller) with horizontal displacement restrained, what would be the support reactions and all member axial forces?
- If the support at point E remained a roller while a new roller support was added at point G, calculate the support reactions and all member axial forces.



Analysis Guide:

- The axial forces in the truss members can be obtained by solving the equilibrium equations at each joint.
- The truss structure, the support conditions, and the loads are all symmetric about the vertical line CG. Therefore, the axial forces in the truss members will also be symmetrically distributed. In this sense, only the equilibrium equations at half of the joints need to be solved.
- For an unloaded joint with three connected members where two are collinear, the third member must be a zero-force member.
- Answers for checking: (a) $v_G = -3228/EA$, $u_E = 900/EA$; (b) $R_{Ex} = -75$; (c) $R_G = 98.76$.

Solution:**Part (a)**

This is a statically determinate truss structure. Support reactions and internal member forces can be calculated directly using global and joint equilibrium equations. Since both the structure and the applied load are symmetric about the central vertical axis (line CG), the internal forces are also symmetrically distributed, as shown below, along with the internal member force distributions due to the unit loads at points G and E.

The displacements at points G and E are computed as follows (the calculations can be performed for only the left half of the structure and then multiplied by two, excluding member GC, which has no contribution anyway):

$$v_G = \sum \frac{nNL}{EA} = 2 \cdot \frac{6}{2EA} \cdot \left(\frac{3}{8} \cdot \frac{225}{4} + \frac{3}{4} \cdot \frac{375}{4} + \frac{-3}{8} \cdot \frac{-225}{4} \right) \quad (140)$$

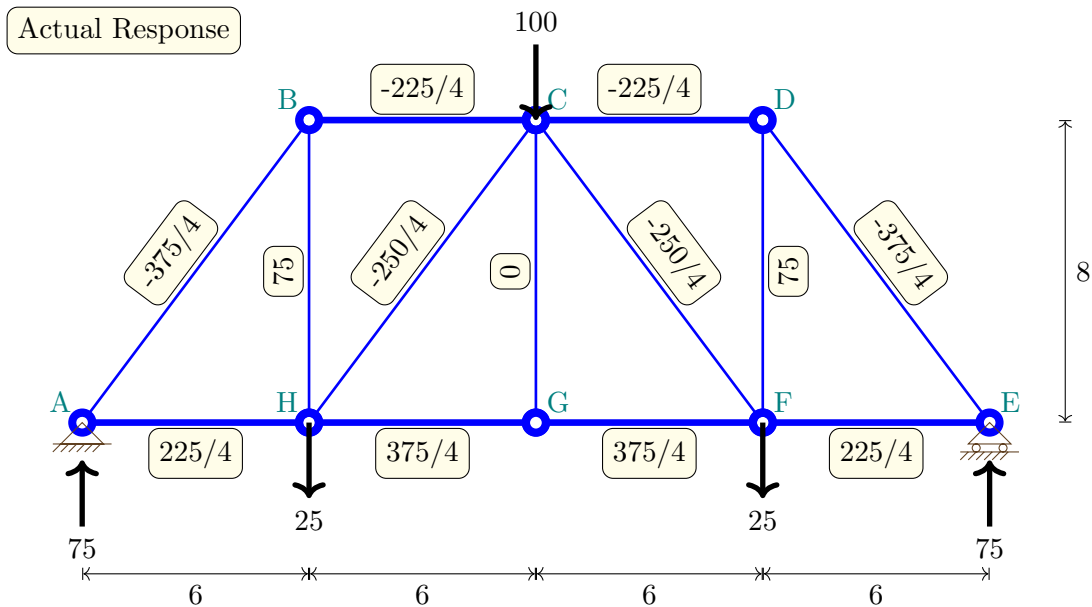
$$+ 2 \cdot \frac{8}{EA} \cdot \left(\frac{1}{2} \cdot 75 \right) + 2 \cdot \frac{10}{EA} \cdot \left(\frac{-5}{8} \cdot \frac{-375}{4} + \frac{-5}{8} \cdot \frac{-250}{4} \right) \quad (141)$$

$$= 3228/EA(\downarrow) \quad (142)$$

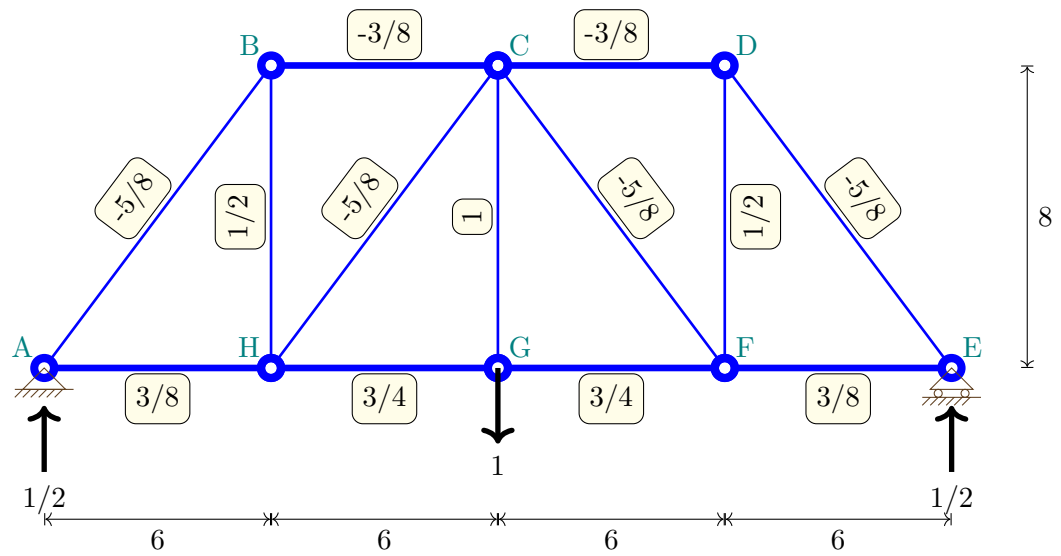
$$u_E = \sum \frac{nNL}{EA} = 2 \cdot \frac{6}{2EA} \cdot \left(1 \cdot \frac{225}{4} + 1 \cdot \frac{375}{4} \right) \quad (143)$$

$$= 900/EA(\rightarrow) \quad (144)$$

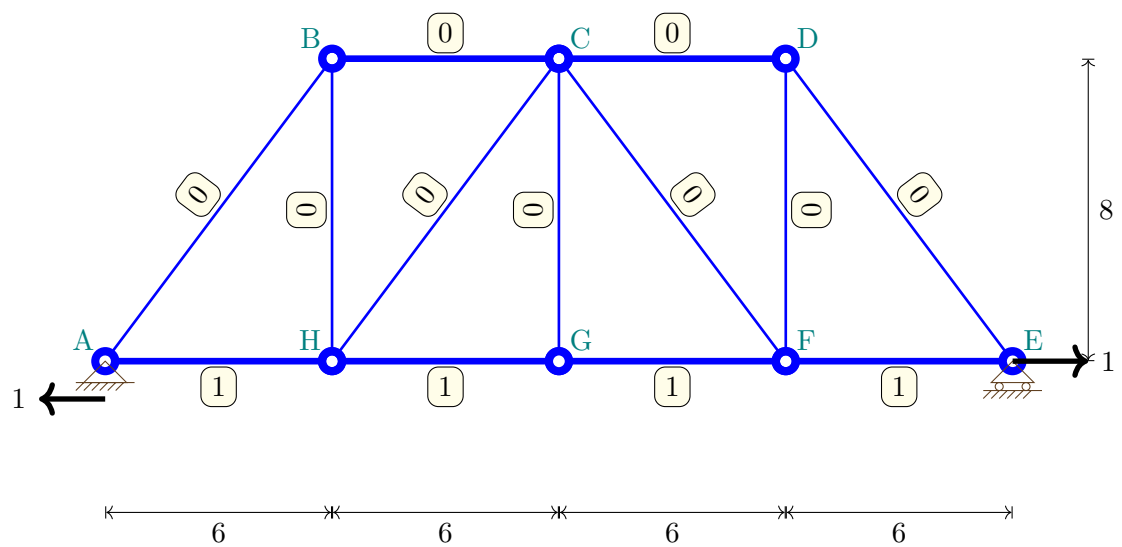
Note that although the value of v_G is positive, it represents a downward displacement because the applied unit load at point G is downward. A positive result confirms that the displacement is in the same direction as the applied unit load.



Unit Load at G



Unit Load at E



Part (b)

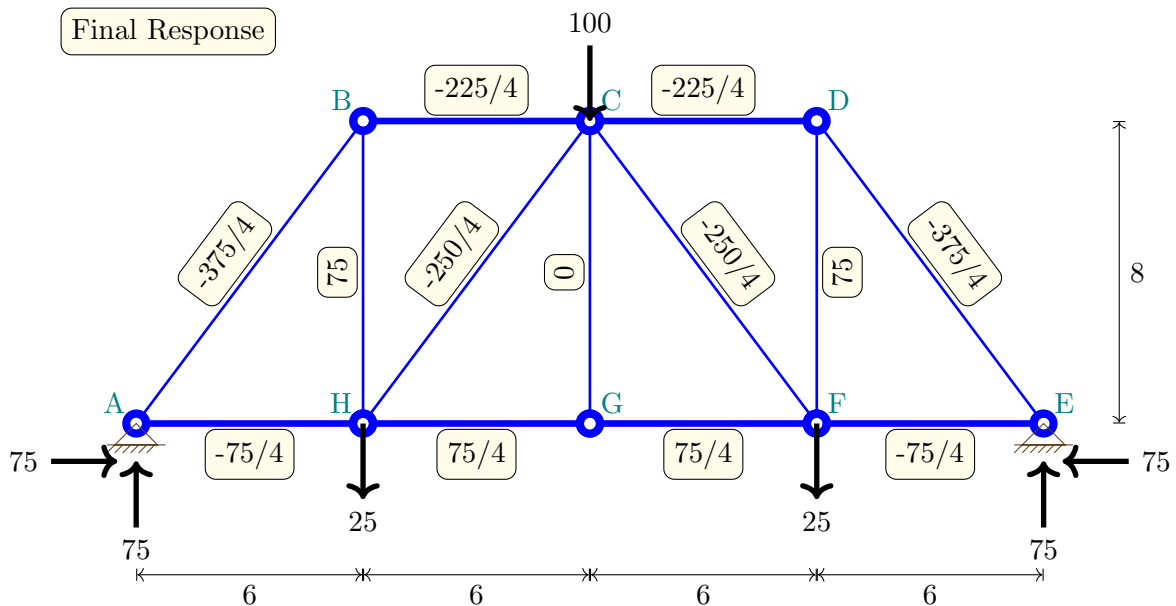
If the support at point E were a pinned support that restrained both horizontal and vertical displacements, we could use the compatibility condition $u_E = 0$ to obtain the support reaction R_{Ex} . Note that we can use the previously obtained results where a unit load was applied at point E.

$$\sum \frac{n^2 L}{EA} = \frac{6}{2EA} \cdot (1^2 + 1^2 + 1^2 + 1^2) = 12/EA \quad (145)$$

$$\Rightarrow R_{Ex} = \frac{-\sum \frac{nNL}{EA}}{\sum \frac{n^2 L}{EA}} \quad (146)$$

$$= -900/12 = -75(\leftarrow) \quad (147)$$

The internal member forces can be computed as $N + n \cdot R_{Ex}$, as shown below:



Part (c)

If the support at point E remained a roller while a new roller support was added at point G, we could use the compatibility condition $v_G = 0$ to obtain the support reaction R_{Gy} . Note that we can use the previously obtained results where a unit load was applied at point G.

$$\sum \frac{n^2 L}{EA} = 2 \cdot \frac{6}{2EA} \cdot \left(\left(\frac{3}{8} \right)^2 + \left(\frac{3}{4} \right)^2 + \left(\frac{-3}{8} \right)^2 \right) \quad (148)$$

$$+ 2 \cdot \frac{8}{EA} \cdot \left(\frac{1}{2} \right)^2 + \frac{8}{EA} \cdot 1^2 + 2 \cdot \frac{10}{EA} \cdot \left(\left(\frac{-5}{8} \right)^2 + \left(\frac{-5}{8} \right)^2 \right) \quad (149)$$

$$= 32.6875/EA \quad (150)$$

$$\Rightarrow R_{Gy} = -3228/32.6875 = -98.76(\uparrow) \quad (151)$$

Note that we have applied a downward unit-load. Therefore, a negative result implies that the reaction at G is upward.

The internal member forces can be computed as $N + n \cdot R_{Gy}$, shown as follows:

